

Mathematical Preliminaries: Metric Spaces

Hyun Jae Stephen Chu*
University of Notre Dame

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1 Metric Space Primer

Previously, we studied the real line $\mathbb{R}$, which carries two important structures:

- **Distance:** $|x - y|$ measures how far apart two points are.
- **Order:** $x < y$ allows us to discuss upper and lower bounds, suprema, and infima.

We now generalize the distance structure. Many mathematical objects are not subsets of $\mathbb{R}$ and often require a more flexible framework. Metric spaces generalize the notion of distance from the real line to arbitrary sets equipped with a distance function, allowing us to extend the results proved for $\mathbb{R}$ to a much broader mathematical setting.

1.1 Metric Spaces and Norms

We first introduce the notion of distance function or **metric** that assigns distances between two points to be measured. Recall that the usual distance between two real numbers is given by $|x - y|$. This distance has several natural properties: it is zero when the two points are equal, it does not depend on the order of the two points, and traveling through an intermediate point cannot shorten the direct distance. A metric retains these essential properties while allowing the underlying set to consist of arbitrary mathematical objects.

Definition 1. Let S be nonempty. A function $d : S \times S \rightarrow \mathbb{R}$ is called a **metric/distance** on S if for all $x, y, z \in S$, the following holds:

1. (Identity) $d(x, y) = 0 \iff x = y$
2. (Symmetry) $d(x, y) = d(y, x)$
3. (Triangle Inequality) $d(x, z) \leq d(x, y) + d(y, z)$

Then the pair (S, d) is called a **metric space**.

*Comments and corrections are welcome and may be sent to hchu@nd.edu.

Example 1. The following are examples of a metric space.

1. The Euclidean space where $S = \mathbb{R}^k$ and $d(x, y) = [\sum_{i=1}^k (x_i - y_i)^2]^{1/2}$
2. The Taxicab space where $S = \mathbb{R}^2$ and $d(x, y) = |x_1 - x_2| + |y_1 - y_2|$
3. The indicator space where for any S and $d(x, y) = \mathbb{1}\{x \neq y\}$

A concept of distance from zero is called the **norm**, which can be formally defined as follows.

Definition 2. A function $\|\cdot\| : \mathbb{R}^k \rightarrow \mathbb{R}$ is called a **norm** on $\mathbb{R}^k$ if for all $x, y \in \mathbb{R}^k$ and $\alpha \in \mathbb{R}$, the following holds:

1. $\|x\| = 0 \iff x = 0$
2. $\|\alpha x\| = |\alpha| \|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Remark 1. Each norm $\|\cdot\|$ on $\mathbb{R}^k$ generates a metric d such that

$$d(x, y) = \|x - y\|$$

Example 2. A p -norm for $p \geq 1$ is

$$\|x\|_p = \left(\sum_{i=1}^k |x_i|^p \right)^{\frac{1}{p}}$$

- if $p = 2$, then the p -norm is the same as the Euclidian norm, i.e.

$$\|x - y\|_2 = \left(\sum_{i=1}^N |x_i - y_i|^2 \right)^{\frac{1}{2}}$$

- if $p = \infty$, then we set

$$\|x\|_{p=\infty} = \max_k |x_k|$$

as $|x_i|^p$ will explode as $p \rightarrow \infty$.

Remark 2. The following holds

$$||x| - |y|| \leq \|x - y\|$$

Proof. Note that

$$\begin{aligned} \|x\| &= \|x - y + y\| \leq \|x - y\| + \|y\| \\ \Rightarrow \|x\| - \|y\| &\leq \|x - y\| \end{aligned}$$

Similarly, we can get

$$\|y\| - \|x\| \leq \|y - x\|$$

Then combining these two, we get

$$|||x| - |y||| \leq ||x - y||$$

□

We can also define a norm and a metric in functions.

Definition 3. Let U be any set and bU be the set of bounded real functions on U such that $f : U \rightarrow \mathbb{R}$. Then the **sup-norm** is

$$||f||_\infty \equiv \sup_{x \in U} |f(x)|$$

This lead to the following metric

$$d_\infty(f, g) \equiv \sup_{x \in U} |f(x) - g(x)|$$

Remark 3. (bU, d_∞) is a metric space.

Proof. Triangle inequality can be shown as follows. Let $f, g, h \in bU$. Then

$$\begin{aligned} |f(x) - g(x)| &= |f(x) - g(x) + h(x) - h(x)| \\ &\leq |f(x) - h(x)| + |h(x) - g(x)| \\ &\leq \sup |f(x) - h(x)| + \sup |h(x) - g(x)| \\ &= d_\infty(f, h) + d_\infty(h, g) \end{aligned}$$

□

Now under this metric space, we will generalize the concepts introduced in a real space.

1.2 Sequences

Definition 4. Let (S, d) be a metric space. A sequence $x_n \in S$ is said to **converge** to $x \in S$ if $\forall \varepsilon > 0 \exists N$ such that

$$d(x_n, x) < \varepsilon \quad \forall n \geq N$$

Definition 5. The ε -**ball** centered on $x \in S$ is given by

$$B_\varepsilon(x) \equiv \{z \in S : d(z, x) < \varepsilon\}$$

Theorem 1. A sequence $\{x_n\} \subset S$ can have at most one limit.

Lemma 1. A sequence $\{x_n\} \subset \mathbb{R}^k$ with d_2 metric converges to $x \in \mathbb{R}^k$ if and only if $x_n^j \rightarrow x^j \in \mathbb{R}$ for all $j = 1, \dots, k$.

Definition 6. Let S, Y be metric spaces and $f : A \rightarrow Y$ where $A \subset S$. The function f is **continuous at** a if $\forall \{x_n\} \subset A$ such that $x_n \rightarrow x$, then we have $f(x_n) \rightarrow f(x) \in Y$. The function is said to be **continuous** if it is continuous at every $x \in A$.

1.3 Sets

Definition 7. A set $A \subset S$ is **open** if $\forall x \in A, \exists \varepsilon > 0$ such that

$$B_\varepsilon(x) \subseteq A$$

A point x is said to be the **interior** of A if $\exists \varepsilon > 0$ such that $B_\varepsilon(x) \subseteq A$. Thus, an open set implies that all points in A is an interior to A .

Definition 8. A set $A \subset S$ is **closed** if $\forall x \in \mathbb{R}$ such that $B_\varepsilon(x) \cap A \neq \emptyset \forall \varepsilon > 0$, then $x \in A$.

Note that a point x is said to **adhere** A if $B_\varepsilon(x) \cap A \neq \emptyset \forall \varepsilon > 0$. Thus, a closed set implies that it contains all points that adhere to A .

Theorem 2. A set $A \subset S$ is closed if and only if for every convergent sequence $\{x_n\} \subset A$, the limit of the sequence is also in A , i.e.

$$[x_n \rightarrow x \text{ and } x_n \in A \forall n] \Rightarrow [x \in A]$$

Theorem 3. A subset of metric space is open if and only if its complement is closed.

Property 1.

- Open sets
 1. Union of open sets are open.
 2. Intersection of finite open sets are open.
 3. The empty set and whole space is open.
- Closed sets
 1. Intersection of closed sets is closed.
 2. Union of finitely many closed sets is closed.
 3. The empty set and whole space is closed.

Remark 4. 1. An intersection of infinite open sets is not open.

2. An union of infinitely many closed sets is not closed.

Theorem 4. A function $f : S \rightarrow Y$ is **continuous** if and only if the preimage $f^{-1}(G)$ of every open set $G \subset Y$ is open in S .

Definition 9. Let (S, d) be a metric space and $A \subset S$.

1. The set A is called **precompact** in S if every sequence in A has a subsequence that converges to a point in S .
2. The set A is called **compact** if every sequence in A has a subsequence that converges to a point in A .

Remark 5. By definition, we can know that

$$\text{compact} \Rightarrow \text{precompact}$$

Definition 10. For a set $K \subset S$, an **open cover** is a collection $\{G_\alpha\}$ of open subsets of S such that $K \subset \cup_\alpha G_\alpha$. The cover is called **finite** if it consists of finitely many sets.

Remark 6. In words, an open cover is a union of open sets that can cover up a set K . This cover is finite if we have a finite number of open sets that cover up K , e.g. K is covered by a union of two open sets.

Theorem 5. A subset K of an arbitrary metric space S is compact if and only if every open cover of K can be reduced to a finite cover.

Recall from the Bolzano-Weierstrass theorem in Remark 1, every bounded sequence in the real space has a convergent subsequence. Combining this with the definition of a closed set, we see a connection to the notions of precompactness and compactness. Let $A \subseteq \mathbb{R}$ be bounded, and consider any sequence $\{x_n\} \subseteq A$. Since A is bounded, the sequence $\{x_n\}$ is also bounded. Therefore, it has a subsequence that converges to some point of the space $\mathbb{R}$.

The limit need not belong to A . If A is also closed, however, then every convergent sequence of points in A has its limit in A . Consequently, every sequence in a closed and bounded set $A \subseteq \mathbb{R}$ has a subsequence that converges to a point of A . This is the sequential characterization of compactness and leads to the Heine–Borel theorem.

Theorem 6. (Heine–Borel Theorem)

A subset $A \subset (\mathbb{R}^k, d_2)$ is **precompact** if and only if it is bounded. Also, it is **compact** if and only if it is closed and bounded.

Remark 7. Note that the above Heine-Borel Theorem holds only in a real space. That is, in a general metric space, one cannot say that a set is compact just because it is closed and bounded. Instead, one should use the direct definition of converging subsequences or Theorem 5 of every open cover begin able to reduced to a finite cover.

1.4 Cauchy Sequences and Complete Spaces

Definition 11. A sequence $\{x_n\} \subset S$ is **Cauchy** if $\forall \varepsilon > 0 \exists N_\varepsilon$ such that

$$d(x_n, x_m) < \varepsilon \text{ for all } n, m \geq N_\varepsilon$$

Property 2. Let (S, d) be a metric space and $\{x_n\}$ be a Cauchy sequence in S .

1. Every Cauchy sequence is bounded, i.e. $\exists M$ such that $d(x_n, x) \leq M \forall n$ where $x \in S$.
2. In any metric space S , every convergent sequence is a Cauchy sequence.
3. In $\mathbb{R}^k$, every Cauchy sequence is convergent to $x \in \mathbb{R}^k$.

Remark 8. By 2 in Property 2 above, we can see that if $x_n \rightarrow x$, then x_n is a Cauchy sequence. Then does the converse hold? As 3 says, it does hold in the real space, but not in a general metric space. Since the converse of 2 does not always hold, we define a metric space where every Cauchy sequence converges to some member in the same space.

Definition 12. A subset A of a metric space S is **complete** if every Cauchy sequence in A converges to some point $a \in A$. If $A = S$, we call S a **complete metric space**.

Example 3. The following are examples of a complete metric space.

1. The set of real numbers under the metric space $(\mathbb{R}, |\cdot|)$.
2. Closed subsets of complete metric spaces are complete. For instance, $(\mathbb{R}^+, |\cdot|)$ is complete, while $(0, \infty)$ is not since sequence $1/n$ is Cauchy but its limit point $0 \notin (0, \infty)$.
3. The Euclidian space $(\mathbb{R}^k, d_2)$ is complete. A sequence $x_n = (x_n^1, \dots, x_n^k)$ in $\mathbb{R}^k$ is Cauchy iff each component sequence x_n^j is Cauchy in $\mathbb{R}$.
4. The space of bounded real-valued functions bU with the sup-norm metric

$$d_\infty(f, g) = \sup_{x \in U} |f(x) - g(x)|$$

is complete.

5. The subspace of continuous bounded functions, denoted as $bcU \subset bU$ with metric d_∞ is also complete.

Theorem 7. Let S be a complete metric space. $A \subset S$ is complete if and only if it is a closed subset of S .

Proof. ($\Rightarrow$) Suppose A is complete and $\{x_n\} \subset A$ with $x_n \rightarrow x \in S$. Since x_n is convergent, it must be Cauchy. Since A is complete, $x \in A$ must hold. Thus, A contains its limit points and is closed.

($\Leftarrow$) Suppose A is closed. Let $\{x_n\}$ be a Cauchy sequence in A . Since S is complete, $\{x_n\} \rightarrow x \in S$. As A is closed, the limit point x must be in A . Thus, A is complete since all Cauchy sequences in A is convergent to a point in A . $\square$

Definition 13. A sequence of functions $f_n : U \rightarrow \mathbb{R}$ **converge pointwise** to f if $|f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$ for all $x \in U$. That is, every single point converges.

A sequence of functions $f_n : U \rightarrow \mathbb{R}$ **converge uniformly** to f if $d_\infty(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$.

Remark 9. *Uniform convergence is stronger than pointwise convergence, i.e.*

$$\text{uniform convergence} \Rightarrow \text{pointwise convergence}$$

Theorem 8. *Let f_n and f be real-valued functions on metric space S . If f_n is continuous on S for all n and f_n uniformly converges to f , i.e. $d_\infty(f_n, f) \rightarrow 0$, then f is also continuous on S .*

2 Optimization and Equivalence

In economics, optimization is important since rational agents solve the optimization problem and act regarding the solutions. Even in statistics and numerical methods, optimization is necessary as one searches for the best estimator in maximum likelihood or seeks a simple representation f_n that is the closest to the function f in some suitable metric sense.

In any optimization problem, the most important thing is that the existence of optima must be guaranteed. If the solution does not exist, there will be no reason to proceed further and try to solve the optimization problem. Discussions of the existence for the solution of optimization typically begins with the following theorem.

Theorem 9. *Let $f : S \rightarrow Y$ where S and Y are metric spaces and f is continuous. If $K \subset S$ is compact, then so is $f(K)$.*

Remark 10. *Theorem 9 above means that functions preserve the compactness of its domain.*

Now given this, the following theorem is one of the most fundamental theorems in optimization theory, since it guarantees the existence of optima. Specifically, it says that for continuous functions on compact domains the optima always exists.

Theorem 10. (Weierstrass Theorem)

Let $f : K \rightarrow \mathbb{R}$ where $K \subset (S, d)$. If f is continuous and K is compact, then f attains its supremum and infimum on K .

Note that given the definitions of maximum and minima, the Weierstrass theorem can be restated so that at least one maximizer and minimizer of a function on compact domain is guaranteed to exist.

Theorem 11. *If K is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then K contains at least one maximizer and minimizer of f on K .*

Note that optimization results like the Weierstrass theorem are fundamentally topological. That is, the results can vary depending on the metric space defined on. However, one notion that guarantees robustness of these results is equivalence.

Definition 14. *Let d and d' be metrics on a nonempty set S . We say that d and d' are **equivalent** if $\exists K, J \in \mathbb{R}^+$ such that $\forall x, y \in S$*

$$d(x, y) \leq Kd'(x, y) \text{ and } d'(x, y) \leq Jd(x, y)$$

Remark 11. *Equivalence is important since equivalent metrics share the same convergent sequences and Cauchy sequences, and equivalent metric spaces (S, d) and (S, d') share the same open, closed, compact, and bounded sets. Therefore, the equivalence of metrics ensure that the optimization results like the Weierstrass theorem are preserved through different metric spaces.*

Lemma 2. *Let d and d' be equivalent on S and $x_n \subset S$. The sequence x_n d -converges to $x \in S$ if and only if it d' -converges to x .*

Proof. Suppose $d(x_n, x) \rightarrow 0$. Then since d and d' are equivalent, we have

$$d'(x_n, x) \leq Jd(x_n, x) \rightarrow 0$$

which implies d' -convergence. Similarly, supposing $d'(x_n, x) \rightarrow 0$, we can get d -convergence. $\square$

Remark 12. *A sequence x_n in a metric space (S, d) is Cauchy if and only if $\lim_{n \rightarrow \infty} \sup_{k \geq n} d(x_n, x_k) = 0$.*

Proof. $(\Rightarrow)$ If x_n is Cauchy, $\forall \varepsilon > 0 \exists N$ such that

$$d(x_n, x_k) < \varepsilon \text{ for all } n, k \geq N$$

Then

$$\sup_{n, k \geq N} d(x_n, x_k) < \varepsilon \text{ for all } n \geq N$$

implying that the limit is 0.

$(\Leftarrow)$ Suppose $\lim_{n \rightarrow \infty} \sup_{n, k \geq N} d(x_n, x_k) = 0$. Then for all $\varepsilon > 0$, there exists N such that

$$\sup d(x_n, x_k) < \varepsilon$$

which implies $d(x_n, x_k) < \varepsilon$ for all $n, k \geq N$. $\square$

Lemma 3. *Let d and d' equivalent on S and $x_n \subset S$. Show that x_n is d -Cauchy if and only if it is d' -Cauchy.*

Lemma 4. *Let d and d' equivalent on S and $A \subset S$. Then A is d -complete if and only if it is d' -complete.*

Lemma 5. *Let d and d' be equivalent on S . Then (S, d) and (S, d') share the same open, closed, bounded, compact sets.*

Lemma 6. *Let d and d' be equivalent on S and let $f : S \rightarrow \mathbb{R} = (\mathbb{R}, |\cdot|)$. Then f is d -continuous if and only if f is d' -continuous.*

Lemma 7. *All metrics on $\mathbb{R}^k$ induced by a norm are equivalent.*

2.1 Fixed Points

Like optimization, fixed point theory has great practical importance. The two theories are closely related because many optimization problems can be reformulated as fixed point problems, often leading to computationally efficient solution methods. We begin by defining a fixed point and then introduce two fundamental fixed point theorems. One uses convexity and was suggested by Brouwer, while the other is a contraction mapping theorem by Banach.

Definition 15. Let $T : S \rightarrow S$ be a self-map. Then $x^* \in S$ is called the **fixed point** of T on S if

$$Tx^* = x^*$$

Remark 13. The map T can be viewed as a transformation that takes an element $x \in S$ and returns another element $Tx \in S$. If S is a space of functions, then T takes a function as its input and returns another function. In this setting T is often called an operator on functions.

Remark 14. Suppose that we wish to find a point $x \in S$ that minimizes $d(Tx, x)$, and that the desired minimum value is zero. Instead of solving this minimization problem directly, we may seek a fixed point of T . Indeed,

$$Tx^* = x^* \Rightarrow d(Tx^*, x^*) = 0$$

Since distances are nonnegative, any fixed point x^* is a global minimizer of $d(Tx, x)$, with minimum value zero. Fixed point theorems can therefore be used to establish the existence of such a minimizer and, in some cases, to provide a method for computing it.

The fixed point condition $Tx^* = x^*$ is equivalent to $d(Tx^*, x^*) = 0$. Thus, a fixed point can also be viewed as a point that minimizes the distance between Tx and x with minimum value zero.

Definition 16. A set $S \subset \mathbb{R}^k$ is called **convex** if $\forall \lambda \in [0, 1]$ and $a, a' \in S$, we have

$$\lambda a + (1 - \lambda)a' \in S$$

Theorem 12. (Brouwer's Fixed Point Theorem)

Consider a metric space $(\mathbb{R}^k, d)$. Let $S \subset \mathbb{R}^k$ and $T : S \rightarrow S$. If T is continuous and S is both compact and convex, then T has at least one fixed point in S .

Definition 17. Let (S, d) be a metric space and $T : S \rightarrow S$.

1. T is called **nonexpansive** on S if $d(Tx, Ty) \leq d(x, y)$ for all $x, y \in S$
2. T is called **contracting** on S if $d(Tx, Ty) < d(x, y)$ for all $x \neq y \in S$
3. T is called **uniformly contracting** on S with modulus $\lambda \in [0, 1)$ if

$$d(Tx, Ty) \leq \lambda d(x, y) \text{ for all } x, y \in S$$

Remark 15. *In plain words, each definition above can be said as follows:*

- 1. If T is nonexpansive, applying T does not increase the distance. Two points may become closer or remain equally far apart.*
- 2. If T is contracting, applying T makes every pair of distinct points strictly closer.*
- 3. If T is uniformly contracting, applying T makes all distances smaller by at least a common proportional amount. There is one number $\lambda < 1$ that works for every pair.*

The difference of contracting and uniformly contracting is that a contracting map always brings distinct points closer but the amount of contraction may depend on the points, while a uniformly contracting map brings every pair closer by a common percentage bound.

Theorem 13. (*Banach's Fixed Point Theorem*)

Let $T : S \rightarrow S$ be a self-map where (S, d) is a complete metric space. If T is a uniform contraction on S with modulus λ , then T has a unique fixed point $x^ \in S$. Also, for all $x \in S$ we have*

$$d(T^n x, x^*) \leq \lambda^n d(x, x^*)$$

and thus $T^n x \rightarrow x^$ as $n \rightarrow \infty$.*

Exercise 1. Let S be the collection of continuous functions on $[0, 1]$. Check if

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

is a valid metric on S .

Exercise 2. Prove or disprove each of the following statements.

(1) An intersection of infinite open sets is open.

(2) A union of infinitely many closed sets is closed.

Exercise 3. Show that if $x_n \leq y_n \leq z_n$ for all $n \in \mathbb{N}$, and if $\lim x_n = \lim z_n = l$, then $\lim y_n = l$ as well.

Exercise 4. Let (a_n) and (b_n) be Cauchy sequences. Decide whether each of the following sequences is a Cauchy sequence, justifying each conclusion.

(1) $c_n = |a_n - b_n|$

(2) $c_n = (-1)^n a_n$